



Graphs in Machine Learning

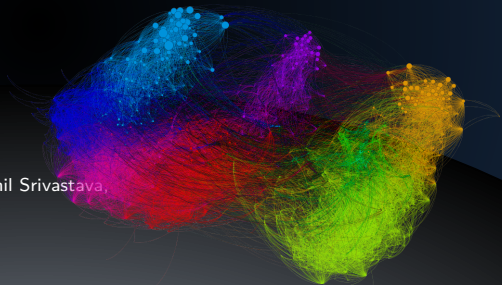
Cut Graph Sparsifiers

Benczur-Karger Algorithm

Michal Valko

Isara Labs

Partially based on material by: Rob Fergus, Nikhil Srivastava,
Yiannis Koutis, Joshua Batson, Daniel Spielman



Cut Graph Sparsifiers

Define G and H are $(1 \pm \varepsilon)$ -**cut similar** when $\forall S$

$$(1 - \varepsilon)\text{cut}_H(S) \leq \text{cut}_G(S) \leq (1 + \varepsilon)\text{cut}_H(S)$$

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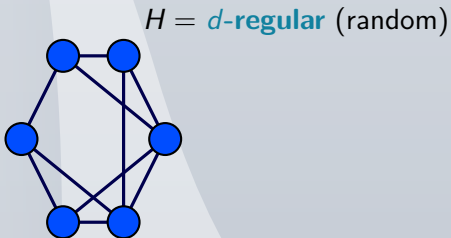
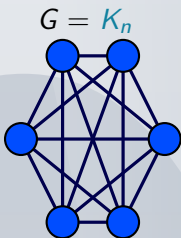
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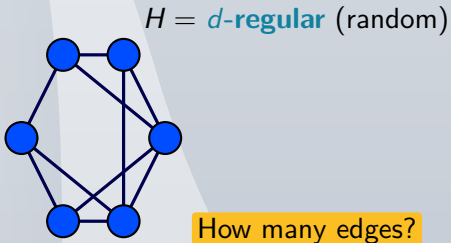
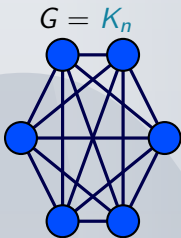
Is this always possible? Benczúr and Karger (1996): Yes!

$\forall \varepsilon \exists (1 + \varepsilon)$ -cut similar H with $\mathcal{O}(n \log n / \varepsilon^2)$ edges s.t. $E_H \subseteq E$
and computable in $\mathcal{O}(m \log^3 n + m \log n / \varepsilon^2)$ time n nodes, m edges

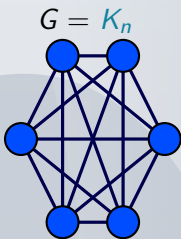
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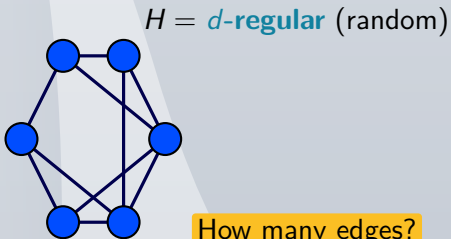
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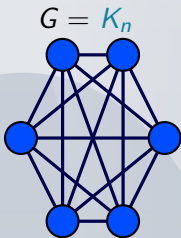


$$|E_G| = \mathcal{O}(n^2)$$

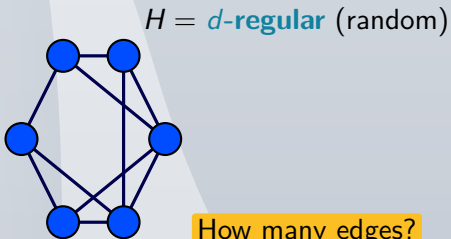


How many edges?

Cut Graph Sparsifiers



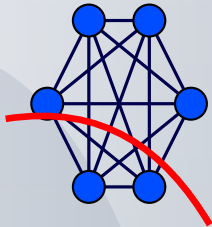
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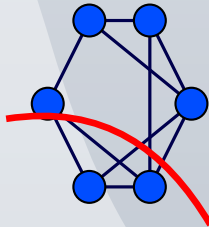
$$|E_H| = \mathcal{O}(dn)$$

Cut Graph Sparsifiers

$$G = K_n$$

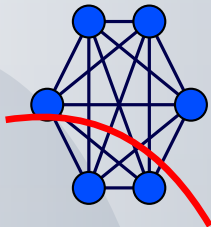


$$H = d\text{-regular (random)}$$

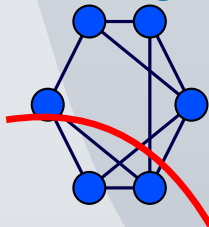


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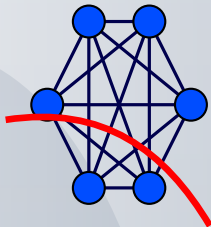
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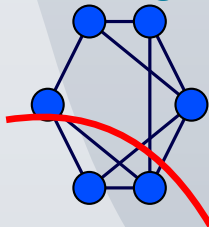
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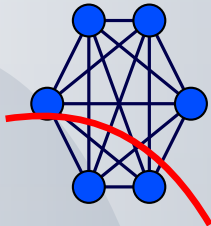


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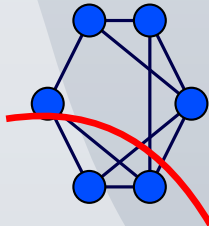
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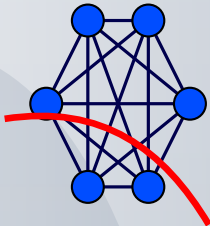
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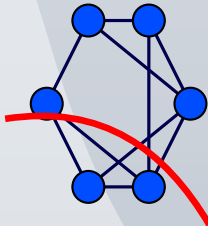
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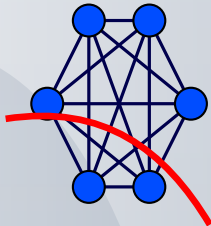
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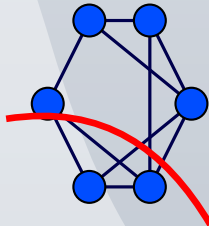
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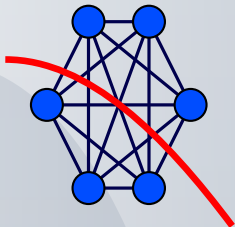
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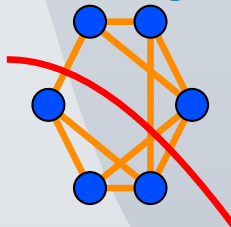
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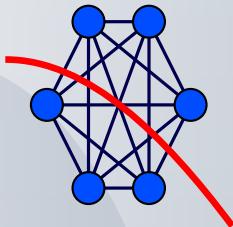


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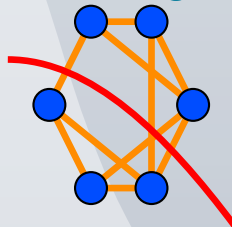


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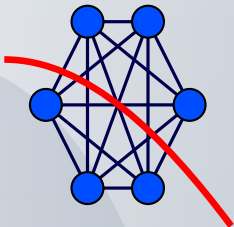
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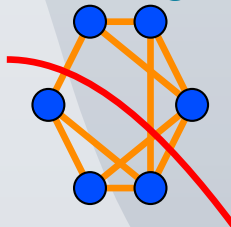
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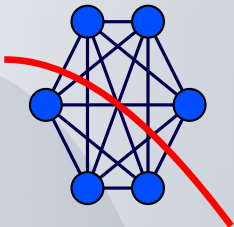


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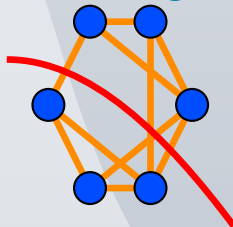
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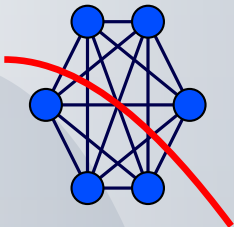
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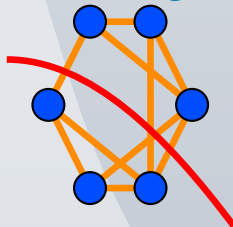
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Benczúr & Karger: Can find such H quickly for any G !

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Recall if $\mathbf{f} \in \{0, 1\}^n$ represents S then $\mathbf{f}^\top \mathbf{L}_G \mathbf{f} =$

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Spectral sparsifiers are stronger!

but checking for spectral similarity is easier

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<https://misovalko.github.io/mva-ml-graphs.html>